The expense ledger

Every tool, service and machine the work runs on. Prices are list prices with the date read, or blank until read; the business share is Christopher's call on each row, and a CPA decides what the return says. Started 2026-11-04 from SOP-ARB-008 (the tech stack, an inventory of the iMac read 2026-11-03), the round-six proposals, `/Applications` as listed on 2026-11-04, and Christopher's own list. Add rows; never erase one, mark it ended. The rules each row is read against are in IRS-NOTES.md.

How to read a row

Column	Means
Kind	subscription (recurring), one-time (bought once), service (usage-billed), free (no charge at the tier in use), open source (no charge, ever, by licence), hardware (lasts more than a year: usually depreciated or expensed under section 179 or the de minimis safe harbor, a CPA's call)
Business share	100% (only the work uses it), mixed (split; the personal part is not deductible), personal, or **?** (not decided)
Owner	who paid and who owns it: Christopher, or a foundation if it bought the thing (then it is their asset, not his expense; proposal 016)
Receipt	where the proof lives (email, App Store, a statement); private paths are named by folder only
Christopher to fill	only he knows the plan or the price; the row stands so the gap is visible, not hidden

A row at **$0** is still a row: the point of the ledger is that nothing the work runs on is missing from it, and a free tool that becomes paid, or that another person would have to buy, is found here first.

Software and services

The fleet and its engines

Item	What it does for the work	Kind	Price (read on)	Billing	Business share	Since	Receipt	Note
GitHub (Free today; **Pro proposed**)	every repository, every lane, every public page, GitHub Pages and the Actions minutes that render the PDFs and run the public-lane sync; Pro adds rules on private repos	subscription	$0 → $4/month (GitHub's upgrade prompt, 2026-11-04; verify on the billing page). Pricing page the same day: Free $0, Team $4 per user per month, Enterprise $21	monthly	100%	2024	GitHub billing emails	001-github-pro.md
Claude (Anthropic)	the fleet's engine: Alfred, Fortuna, Mystarch, the workflows; Claude Code CLI 2.1.317 on 2026-11-03	subscription	plan and price: Christopher to fill	monthly	? (the fleet is the work; some personal use)	2026	Anthropic billing	usage limits hit twice on Nov 4
Littlebird 0.87.7	the observer: screen context, fleet memory, meeting notes; also the `littlebird` MCP server	subscription	Christopher to fill	monthly	?	2026	Littlebird billing	
Firecrawl	web capture for the fleet; the `firecrawl` MCP server	subscription	plan: Christopher to fill; credits reset 2026-11-12	monthly	100%	2026	Firecrawl billing	out of credits this week

Harness Cosmos (formerly Augment Cosmos)	cloud agents	planned; no active subscription	$0 now; price when subscribed: Christopher to fill		100%	planned		Harness bought select Augment assets on 2026-10-08 (SOP-ARB-008)
Augment Intent 0.5.0 / Auggie CLI	Kavanah's old seat; the app is still installed, the `auggie` MCP server does not connect	ended	$0 now					keep the row: history
21st-dev (MCP server)	*UI components*, by its name; not authenticated on 2026-11-03	not in use	$0 as it stands; paid plans exist: Christopher to fill if one is taken		?			reconnect or drop it (SOP-ARB-008, yours to do)
rtk 0.29.0	a proxy to cut agents' token use (its own description)	installed via Homebrew	$0 as installed; licence not checked		100%	2026		use not checked
ntfy	phone alerts for the desks (SOP-ARB-005)	free tier	$0		100%	2026		a paid tier exists if volume grows
shields.io badges, GitHub Pages	the badges on every page; the public twins' hosting	free	$0		100%			

Development tools

Item	What it does for the work	Kind	Price (read on)	Billing	Business share	Since	Receipt	Note
Visual Studio Code 1.99.3	the main editor	free	$0		100%			runs from `~/Downloads`; move it to `/Applications` (SOP-ARB-008)
Sublime Text (build 4215)	a second editor	evaluation or licence	a paid licence exists for continued use: Christopher to fill whether one was bought	one-time	100%			lives in `~/Downloads`
Xcode 16.4	Apple's tools and compilers	free with a Mac	$0		100%			
Terminal, Safari, Keynote, Numbers, Pages, GarageBand, iMovie	Apple's own apps on the iMac; Numbers and Pages opened this month	free with a Mac	$0		mixed			Numbers and Pages are where tables and letters get drafted

GitHub Desktop 3.6.5	a picture of git for people	free	$0		100%			not checked
Docker Desktop 4.66.1 (Docker 29.3.1)	containers; carries its own Linux VM	free at Docker's Personal tier for individuals and small businesses (Docker's own subscription terms; verify on its pricing page on the day)	$0		100%			engine off at the Nov 3 sample
Homebrew 7.0.11	installs the command-line tools below	open source	$0		100%			
Node (v20 through v24 via nvm), npm 10.9.9, pnpm 10.33.0, corepack	the bot, the dashboard, the PDF renderer, mermaid-cli, md-to-pdf	open source	$0		100%			
Python 3.14.3, uv 0.11.2	scripts and the public-lane tripwire	open source	$0		100%			
Rust 1.98.1	builds	open source	$0		100%			not checked
Hardhat 2.24.2, solc (five cached compilers)	the DEX bot's contracts	open source	$0		100%	2025		five compilers need Rosetta (SOP-ARB-006)
git 2.39.5 (Apple), gh 2.103.3, git-filter-repo 2.47.0	version control; the public-lane sync	open source	$0		100%			
ggshield 1.49.0, pre-commit 4.2.0	catch a secret before it is committed	free tier (GitGuardian) and open source	$0; GitGuardian's paid tiers exist: Christopher to fill if one is taken		100%			hooks vary by repo
poppler 26.04.0, pngquant 3.0.3	PDF text and image helpers; the tripwire reads PDFs with pdftotext	open source	$0		100%			
gcloud CLI 587.0.0	Google Cloud from the command line	free CLI; cloud usage is billed per project	$0 for the CLI; any project's usage: Christopher to fill	monthly if any	?		Google Cloud billing, if any	not checked
googleworkspace-cli (gws) 0.22.5	Google Workspace from the command line	free CLI	$0		?			the Workspace plan itself is a row below

ngrok 3.36.1	tunnels	free tier; paid plans exist	$0 as far as known: Christopher to fill if a plan is held	monthly if any	100%			not checked
Rosetta, FileVault, Time Machine, launchd, Universal Control	Apple's translation layer, disk encryption, backup, scheduled agents, shared keyboard	free with macOS	$0		mixed			no Time Machine destination is configured today (SOP-ARB-006)
OpenCore Legacy Patcher	keeps the 2014 MacBook Pro on a current macOS	open source	$0		mixed			unsupported by Apple
VMware Fusion Pro	Windows and Linux on the Macs (proposal 013)	free for all use since Nov 11, 2024 (VMware, read 2026-11-03)	$0		100%	planned		needs a Broadcom account; not installed
Parallels Desktop	used in the past; proposal 013's first choice if its test passes	ended; may return	Christopher to fill if revived	yearly	100%			not installed
Windows 11 (guest licence)	Windows-only trading tools, when one is named	planned	Christopher to fill; Microsoft requires "a unique license... for each instance" (proposal 013)	one-time per VM	100%	planned		the licence question is step 4 of 013's test
Autodesk Fusion 2603.1.15	design and machining software in `~/Applications`; no part of the work	personal-use licence as far as known	Christopher to fill if a paid plan		personal			the name trap: nothing to do with VMware Fusion

Trading

Item	What it does for the work	Kind	Price (read on)	Billing	Business share	Since	Receipt	Note
TradingView 3.5.0	charts and alerts for the trading desks; the `tradingview` and `tv-alerts` MCP servers read it	subscription	plan: Christopher to fill	monthly or yearly	? (trading is the work)		TradingView billing	a trader's data and charting tool is a classic "ordinary" cost in that trade; a CPA confirms
Tradovate Trader 3.260904.0 (platform, data, routing); the `tradovate` MCP server	futures	service	fees per statement	monthly	?		broker statements (private lane)	Intel only, needs Rosetta; gone at macOS 28
Robinhood; the `robinhood` MCP server	stocks and options; market	service	$0 to hold; fees and	per statement	?		broker statements	

	reads for the fleet		interest per statement				(private lane)	
Prop-firm evaluations and resets (TakeProfitTrader, TopOne, MyFundedPerps); the `myfundedperps` MCP server	the funded-account path	one-time per evaluation	per receipt	per event	?	2026	`data/imports` in trading-assistant (redacted receipts, public on purpose)	whether an evaluation fee is a deductible trading expense or a personal cost is exactly the kind of question for a CPA
TradeCopia (as Christopher named it; product and fee to verify)	copy-trading	subscription	Christopher to fill		?			verify the product's exact name and what it is before the row stands
TradeZella to STB 1.3	an Automator helper for trade records; exact job unverified	the lab's own script	$0		100%			if a TradeZella subscription sits behind it, that is its own row: Christopher to fill
Hummingbot; the `hummingbot-mcp` server	market-making bot; clones and an MCP server exist	open source	$0		100%			not checked
Exchange and network fees (BTCC, DEXes, gas)	the crypto and DEX work	service	per trade	per trade	?		exchange exports (private lane)	fees are usually part of cost basis or trading expense; a CPA decides
RPC provider (Alchemy)	the chain reads behind the DEX edge monitor	free tier as far as known	$0; Christopher to confirm the plan	monthly if any	100%	2026		the VPN resolver drops it now and then (SOP-ARB-002)
The DEX edge monitor	this lab's own code: three launchd agents on the iMac	the lab's own	$0		100%	2026		monitor-only; funding gated on the edge study
`astra-native-main`, `astra-native-service` (MCP servers)	purpose unverified	not checked	$0 as far as known		?			named in the trading-assistant project
Two MCP servers named for an organisation's domain	an organisation's tools, not this ledger's	not this ledger's						counted in SOP-ARB-008, not named

Communications

Item	What it does for the work	Kind	Price (read on)	Billing	Business share	Since	Receipt	Note
Discord 0.0.420	the fleet's rooms and the	free	$0		mixed			

	organisations' servers							
Discord Nitro	the main Discord account's upgrade	subscription	Christopher to fill	monthly	mixed (fleet rooms and personal)			
Telegram 13.0, WhatsApp 26.38.74	people and alerts	free	$0		mixed			Telegram Premium, if held, is its own row: Christopher to fill
Slack 4.52.178	work rooms	free tier unless a workspace is paid	$0 as far as known: Christopher to fill if a workspace is paid		?			
Zoom 7.1.9	calls	free tier unless a plan is held	$0 as far as known: Christopher to fill if a plan is held		?			
Keet 2.5.2	peer-to-peer calls	free	$0		?			not checked
Google Workspace / Business (if any)	mail and documents	subscription	Christopher to fill		?			THEF's own plan is Kenney's, not this ledger's
Notion 7.40.0	notes and pages	free Personal plan unless upgraded	$0 as far as known: Christopher to fill if a paid plan		?			open at the Nov 3 sample
Mobile carrier plan (the iPhone's line)	the phone the desks ring; the hotspot when Starlink drops	subscription	Christopher to fill	monthly	mixed		carrier billing	a phone plan's business share is a split (Publication 334)
Starlink	the connection the monitors and sessions run on	subscription	Christopher to fill	monthly	mixed		Starlink billing	the business share of a home connection is a split, like a phone

Security, backup and storage

Item	What it does for the work	Kind	Price (read on)	Billing	Business share	Since	Receipt	Note
1Password 8.12.42 (and 1Password for Safari)	the vault: credentials, the Apple recovery key	subscription	Christopher to fill	yearly	mixed		1Password billing	load-bearing since Nov 4 (SOP-COM-002)
NordVPN 10.12.0	the VPN the desks run behind (SOP-ARB-002 has its resolver trouble)	subscription	Christopher to fill	yearly or multi-year	mixed		NordVPN billing	Christopher named "Nord" on Nov 4
Intego: VirusBarrier 10.9.101, NetBarrier 10.9.38, ContentBarrier	antivirus, firewall, content filter, updates,	subscription	Christopher to fill	yearly	mixed		Intego billing	NetBarrier's firewall

10.9.28, NetUpdate 10.9.38, Washing Machine 10.9.27	cleanup on the iMac							extension is active
Intego Personal Backup 10.9.29	Intego's own backup, part of the same bundle	included above	$0 beyond the bundle		mixed			whether any job is set up in it is unverified; its use was declined in 015
iCloud+ (2 TB)	the git-safety bundles, phone backup, shared folders	subscription	$9.99/month (Apple's US price; verify on the day)	monthly	mixed		Apple receipts	the 710 MB of bundles is a sliver of it
AppleCare (if any)	cover for the iMac or the iPhone	subscription or one-time	Christopher to fill if held		mixed		Apple receipts	

Sites, names and everyday apps

Item	What it does for the work	Kind	Price (read on)	Billing	Business share	Since	Receipt	Note
Wix	the earlier venture's site, still up	subscription	Christopher to fill	yearly	? (a prior business; personal unless revived)			
Domains	names for the sites	subscription	per domain	yearly	?		registrar emails	list each domain in its own row when known
Brave 155, Firefox 144, Google Chrome 156	browsers; Chrome is also the engine the local PDF renderer uses	free	$0		mixed			
Spotify 1.3.4	music	subscription	Christopher to fill	monthly	personal			not the work's
Capital One Shopping, PayPal Honey, Desktop Verse, Memory Diag	browser helpers, a verse on the desktop, a memory checker	free as far as the inventory shows	$0		personal			listed so `/Applications` has no gap
DappU coursework	the Web3 engineering training behind the dappu/ repos	one-time	Christopher to fill		?	2025–2026		education that maintains or improves skills in a present trade is a recognised category; a new trade is not; a CPA draws the line

Hardware

Item	What it does for the work	Kind	Price	Owner	Business share	Since	Note
iMac 24" (M4, 32 GB, 2 TB)	the workstation everything runs on today	hardware	Christopher to fill	Christopher	mixed	2025	depreciation or section 179 or de

	(SOP-ARB-006); macOS 15.4.1 on 2026-11-03						minimis, by price: a CPA's call
MacBook Pro 2014 (Intel)	the second screen and call machine (proposal 012); a patched macOS	hardware	long since paid	Christopher	mixed	2014	battery and OS work planned (012)
iPhone 14 Pro Max (1 TB)	alerts, ntfy, the phone the desks ring	hardware	Christopher to fill	Christopher	mixed		
iPad Pro	named in SOP-ARB-008 beside the three Macs; its part in the work: Christopher to say	hardware	Christopher to fill	Christopher	?		Universal Control caps at three devices
Mac Studio M5 Max (64 GB / 1 TB, the lean)	the foundation's workhorse	hardware	$3,799 (Apple, read 2026-11-03; proposal 018)	**THEF, if it buys**	not Christopher's expense if so	planned	a financed machine is the buyer's asset (016)
ASUS ProArt PA27JCV 5K	the Studio's display	hardware	$544.95–$569.95 used (B&H, 2026-11-03)	THEF, if it buys	as above	planned	
WD My Passport Ultra 6 TB	Time Machine for three Macs (proposal 019)	hardware	$179.99 (2026-11-03, verify at checkout)	THEF or Christopher	?	planned	under the de minimis amount either way
Secretlab MAGNUS Evo desk, TITAN Evo XL chair	the desk the workstation sits on (proposal 014)	hardware	$749 / verify XL	Christopher unless offered	?	wished	014 says: outside THEF's ask unless they offer
UPS (proposal 015)	keeps the monitors up through a dip	hardware	per 015's table	?	?	planned	
Starlink dish and router	the connection's hardware	hardware	Christopher to fill (bought or leased with the plan)	Christopher	mixed		
Nebra outdoor miner (Helium)	a past DePIN experiment	hardware	paid	Christopher	ended		the frontier had its pilgrims (011 addendum)

The home office (a CPA question, recorded for when it is asked)

The simplified method is $5 per square foot of space used "regularly and exclusively" for the trade, up to 300 square feet (Publication 334, 2025). The regular method (Form 8829) would instead take a share of the home's actual costs, electricity among them; the workstation runs all day, so that share is real but unmeasured: Christopher to fill if the regular method is ever weighed. The desk's corner would have to be measured and used only for the work. Nothing is claimed by this page; it only keeps the facts ready.

What it would take to replace Christopher

The honest account Christopher asked for on Nov 4: the seats, subscriptions and machines another person would need to run the same desks for the organisations this lab serves through its tech work. The ledger above is the detail; this is the sum, by desk. Where a price says "Christopher to fill", the sum does too.

Desk	What another person would need	Seats and subscriptions	Machines	Monthly, where known (read 2026-11-04)

The fleet (Alfred, Fortuna, Mystarch; every public page and sync)	a Claude seat, a GitHub account with Pro, Littlebird, Firecrawl, a secrets scanner	Claude (Christopher to fill), GitHub Pro $4, Littlebird (fill), Firecrawl (fill), GitGuardian free tier	the workstation below	$4 + three plans to fill
The trading desks (futures, stocks, prop firms, copy-trading)	a charting seat, a futures platform with data, a brokerage, the evaluations as they come	TradingView (fill), Tradovate data and routing (per statement), Robinhood ($0 to hold), prop-firm evaluations (per event), TradeCopia (fill)	the workstation; a second screen	per statement and per event; two plans to fill
The Web3 desk (the DEX edge monitor, the dappu repos)	an RPC provider, the open-source toolchain, the coursework behind it	Alchemy (free tier as known), Hardhat and Node $0, DappU (fill, one-time)	the workstation, running all day	$0 known; the coursework one-time
The organisations' tech work (THEF, PIR, O-R-G, Psanctuary, the Church, Taylor'd, the heal site)	the same GitHub account, their reading rooms (free), Google Workspace where one is held, Discord	GitHub (above), Workspace (fill, if any), Discord Nitro (fill)	the workstation	included above; two to fill
Communications	phone line, connection, VPN, the rooms	mobile carrier (fill), Starlink (fill), NordVPN (fill), Slack and Zoom free tiers	the iPhone	three plans to fill
Security and backup	a password vault, endpoint protection, a backup destination, cloud storage	1Password (fill), Intego (fill), iCloud+ $9.99	a Time Machine drive ($179.99, proposal 019)	$9.99 + two plans to fill
The workstation itself	an Apple-silicon Mac with 32 GB or more, a phone, a second machine for calls and the Intel tools	macOS and its tools $0	iMac M4 (fill), MacBook Pro 2014 (paid), iPhone 14 Pro Max (fill); or the Mac Studio at $3,799 if THEF buys	hardware, once; a CPA decides how it is recovered
The hours	the part no subscription replaces: the days at the desks, the nights the monitors are watched, the reading behind every page			Christopher to fill; the ledger counts tools, and this line is the reason the tools are counted

Known monthly subscriptions today, from the rows with a price: GitHub Pro $4 (proposed) + iCloud+ $9.99 = **$13.99**, with every other plan still "Christopher to fill". That gap is the next thing this page closes; see below.

What to add next

- The exact plans and prices for Claude, Littlebird, Firecrawl, TradingView, Starlink, the mobile carrier, Intego, 1Password, NordVPN, Wix, Nitro, Spotify.
- Each domain as a row.
- The receipts' folders, named by folder only.
- A yearly "read on" refresh of every price, and the ended rows marked.
- The hours, however Christopher chooses to count them.

Not tax advice. Kept so the truth is cheap to tell.